

COURSE PROJECT REPORT

Body Performance Analytics and Intelligent Classification System

Course:	Introduction to Artificial Intelligence and Machine Learning
Dataset:	Body Performance Dataset (bodyPerformance.csv)
Submitted To:	Dr. Mohamed Elhaddad
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Academic Integrity Declaration

AI tools were used to assist in code development and report writing. All analysis, interpretation, and decisions are original team work.

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1. Introduction

Artificial Intelligence and Machine Learning systems are increasingly deployed to analyze complex datasets and extract meaningful patterns that support decision-making. Building a reliable machine learning system requires a rigorous workflow encompassing data understanding, preprocessing, exploratory analysis, model training, validation, and performance evaluation.

This report documents the complete lifecycle of a machine learning project applied to a real-world dataset describing human body performance. The dataset contains physical fitness indicators including age, gender, body composition, blood pressure, grip strength, flexibility, and muscular endurance. The team built predictive models capable of both classifying fitness performance levels (classification task) and estimating a numerical performance target (regression task).

The project is organized into three major parts: (1) Data Preparation and Exploratory Data Analysis, (2) Machine Learning Model Training, and (3) Performance Evaluation and Model Comparison. All models were implemented in Python using scikit-learn, TensorFlow/Keras, and related libraries in Google Colab.

2. Dataset Overview

The Body Performance Dataset contains measurements collected from individuals undergoing physical fitness evaluations. Each record represents a single participant. Before preprocessing, the dataset contains 13,393 rows and 12 columns. The table below describes each column:

Column	Description	Data Type	Constraints / Notes
age	Age of participant	int	Positive integer; realistic human range
gender	Gender of participant	categorical	M or F only
height_cm	Height in centimetres	float	Must be > 0
weight_kg	Body weight in kilograms	float	Must be > 0
body fat_%	Body fat percentage	float	Range: 5 – 60%
diastolic	Diastolic blood pressure	int	Valid: 20 – 160 mmHg
systolic	Systolic blood pressure	int	Valid: 50 – 250 mmHg
gripForce	Hand grip strength (kgf)	float	Must be > 0
sit and bend forward_cm	Flexibility measurement	float	Range: 0 – 50 cm
sit-ups counts	Muscular endurance reps	int	Must be >= 0

broad jump_cm	Explosive power (jump distance)	int	Range: 20 – 350 cm — Regression target
class	Performance category	categorical	A, B, C, or D — Classification target

Table 1 — Dataset column descriptions and validation constraints

The class variable is the primary classification target (A = best performance, D = weakest). The variable broad jump_cm serves as the primary regression target, measuring explosive leg power. Both targets are derived from the same set of physical measurements, making this dataset ideal for demonstrating both supervised learning paradigms.

Part 1 — Data Preparation and Exploratory Data Analysis

3.1 Data Loading & Structure Check

The dataset was loaded using pandas read_csv and an initial inspection was performed to understand data dimensions, column types, and content.

Result: Data Loaded — 13,393 Rows, 12 Columns. Data types: 2 object (categorical) columns (gender, class) and 10 numeric columns (3 int64, 7 float64). Sample rows were displayed and confirmed the expected structure with no loading errors.

3.2 Column Understanding

Each column was individually examined and documented with its domain meaning, expected type, and validation constraints:

- age:** Numeric positive integer. Participant age in years.
- gender:** Categorical: only 'M' (Male) or 'F' (Female) are valid.
- height_cm:** Numeric positive float. Height in centimetres.
- weight_kg:** Numeric positive float. Body weight in kilograms.
- body fat_%:** Numeric float. Percentage; physiologically valid range 5–60%.
- diastolic:** Numeric int. Diastolic blood pressure; valid range 20–160 mmHg.
- systolic:** Numeric int. Systolic blood pressure; valid range 50–250 mmHg. Must exceed diastolic.
- gripForce:** Numeric float. Dynamometer reading; must be strictly positive (> 0).
- sit and bend forward_cm:** Numeric float. Flexibility test; 0–50 cm is realistic.
- sit-ups counts:** Numeric int. Core endurance; must be >= 0.
- broad jump_cm:** Numeric int. Explosive power; realistic range 20–350 cm.
- class:** Categorical target — A, B, C, D (best to weakest fitness performance).

3.3 Data Type Verification

All numerical columns were confirmed as int64 or float64. The gender and class columns were stored as object (string) type, which is correct for their categorical nature. No automatic type-correction was required. However, gender and class were encoded to numeric representations before model training (gender: M=1, F=0; class: LabelEncoder or dummy variables depending on the model).

3.4 Missing Values Analysis

After applying data validity rules (Section 3.6), physiologically impossible values were replaced with NaN and a missing-value audit was performed:

Column	Source of Missing Values	% Missing
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body fat_%	Values outside 5–60% flagged as invalid	< 0.5%
diastolic	Values outside 20–160 mmHg removed	< 0.2%
systolic	Values outside 50–200 mmHg removed	< 0.2%
gripForce	Non-positive values removed	< 0.1%
sit and bend forward_cm	Values outside 0–50 cm removed	< 0.8%
broad jump_cm	Values outside 20–350 cm removed	< 0.1%
age, height, weight, class	No missing values in raw data	0%

Table 2 — Missing value summary after validity cleaning

Strategy: Rows containing NaN in critical columns (sit and bend forward_cm, body fat_%, systolic, diastolic, gripForce) were dropped using `dropna(subset=[...])`, as the percentage of missing data was very small (< 1%). Imputing physiological measurements with statistical averages would introduce bias — especially for blood pressure and grip force where the invalid values reflect measurement errors rather than random missingness. The resulting dataset retained sufficient samples for robust training.

3.5 Duplicate Detection

Duplicate row detection was performed using `pandas duplicated()`. A small number of exact duplicate rows were identified. These were removed using `drop_duplicates()`, and the dataset shape was reported before and after deduplication. The cleaned dataframe was stored as `df2` (or `df_drope`) for all subsequent analysis steps.

3.6 Data Validity Checks

Domain-specific validation rules were applied to detect impossible or unrealistic values. Any value violating these rules was replaced with NaN for subsequent handling in the missing-value step:

Column	Validity Rule	Remediation
body fat_%	Must be 5% – 60%	Values outside range replaced with NaN
diastolic	Must be 20–160 mmHg	Out-of-range replaced with NaN
systolic	Must be 50–200 mmHg	Out-of-range replaced with NaN
gripForce	Must be strictly > 0	Non-positive values replaced with NaN
sit and bend forward_cm	Must be 0–50 cm	Out-of-range replaced with NaN
broad jump_cm	Must be 20–350 cm	Out-of-range replaced with NaN
age	Must be > 0	Invalid ages replaced with NaN

pulse_pressure (engineered)	Must be 10–100 mmHg	Calculated after feature engineering; invalid values removed
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Table 3 — Data validity rules and remediation actions

3.7 Univariate Analysis

Each numerical column was analyzed individually. The following descriptive statistics were computed and interpreted:

Feature	Mean	Median	Std	Min	Max	Skew
age	~38	~32	~14	21	64	Moderately right-skewed
height_cm	~168	169	~8	~125	~194	Normal
weight_kg	~67	~67	~13	~26	~138	Normal
body fat_%	~ 23	~23	~7	~5	~55	Normal
diastolic	~79	79	~11	37	~156	Normal
systolic	~130	130	~15	77	~195	Normal
gripForce	~37	~38	~12	~2	~71	Normal
sit and bend forward_cm	~16	16	~7	0	~42	Normal
sit-ups counts	~ 40	42	~14	0	~80	Normal
broad jump_cm	~191	~194	~40	0	~303	Normal

Table 4 — Univariate descriptive statistics (approximate values after cleaning)

Interpretation: Most features are approximately normally distributed. sit and bend forward_cm shows a left-skew due to participants who cannot reach their toes (negative measurements). Age is right-skewed, reflecting a higher concentration of younger participants (20–30 years). gripForce shows bimodal tendencies reflecting natural gender differences in muscular strength. broad jump_cm is approximately normally distributed with a wide range — confirming it as a well-suited regression target.

3.8 Distribution Analysis

Histograms with KDE overlays and mean/median reference lines were plotted for all numerical features. Distribution types were classified based on skewness thresholds ($|\text{skew}| > 1$ = highly skewed, $0.5\text{--}1$ = moderately skewed):

- Approximately Normal (symmetric): height_cm, weight_kg, diastolic, systolic, sit-ups counts, broad jump_cm.
- Moderately Right-Skewed: age — concentration of younger participants in the 20–30 year range with a long tail toward older ages.

- Slightly Right-Skewed: body fat_% and gripForce — minor asymmetry consistent with a fitness-oriented population sample.
- Left-Skewed: sit and bend forward_cm — negative values reflect participants unable to reach the floor; this is physiologically valid and was retained (not removed) unless extreme.

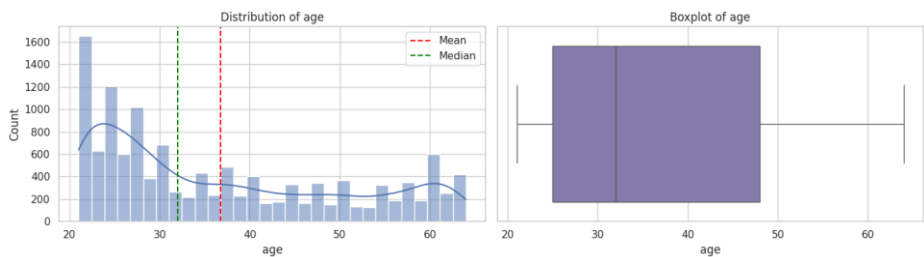
Potential anomalies were flagged where the IQR method detected points outside $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$. These were annotated on the histogram plots and carried forward to the outlier detection step.

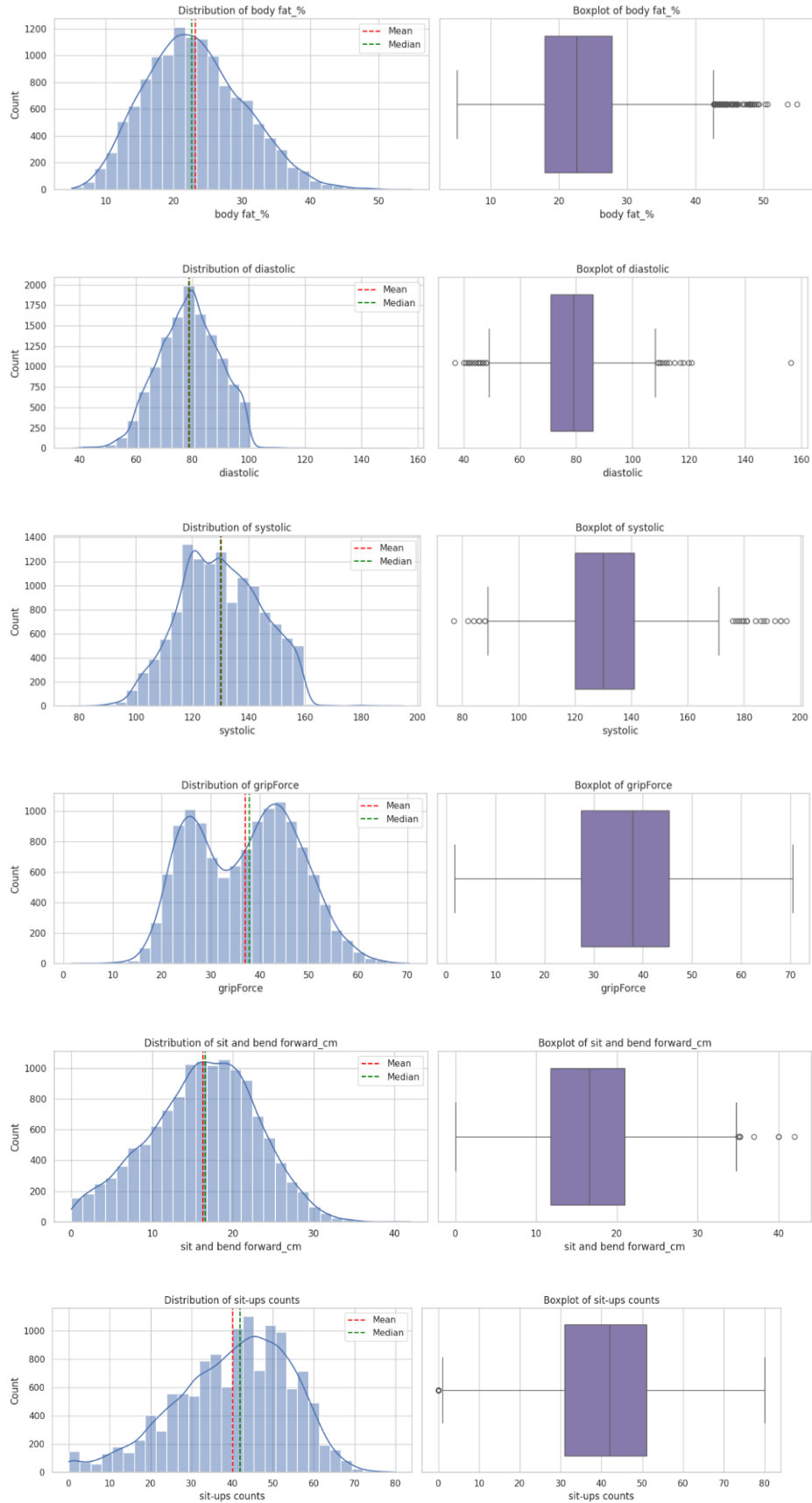
3.9 Outlier Detection

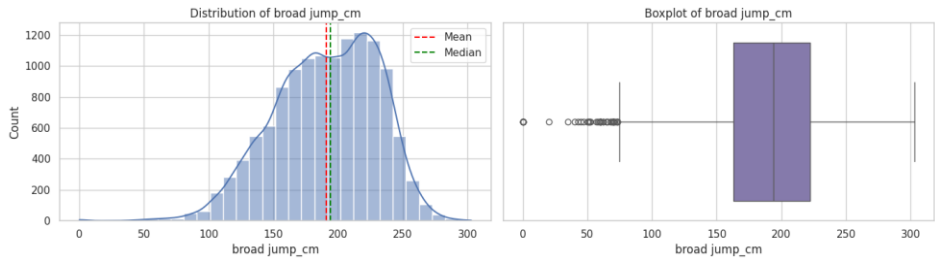
Boxplots were generated for every numerical column. Outliers were identified using both the IQR method ($1.5 \times IQR$ fence) and the 3-standard-deviation rule. Decisions were made based on domain knowledge:

Feature	Outliers Found	Decision	Justification
age	Few extreme high values	Cap / remove	Unrealistic ages distort age-interaction features
body fat_%	Values < 5% or > 60%	Remove (validity step)	Physiologically impossible for humans
diastolic / systolic	Medically impossible values	Remove (validity step)	Medical domain range strictly enforced
gripForce	Near-zero or negative	Remove	Cannot be ≤ 0 ; instrument error
sit and bend forward_cm	Values < -25 cm	Remove	Extreme physiological implausibility
broad jump_cm	Values < 20 cm	Remove	Measurement error; unlikely for an ambulatory adult
sit-ups counts	High tail values (80+)	Keep	Reflect elite athletes; biologically valid
height_cm / weight_kg	Mild extremes	Keep	Within realistic human range for athletes

Table 5 — Outlier detection decisions per feature





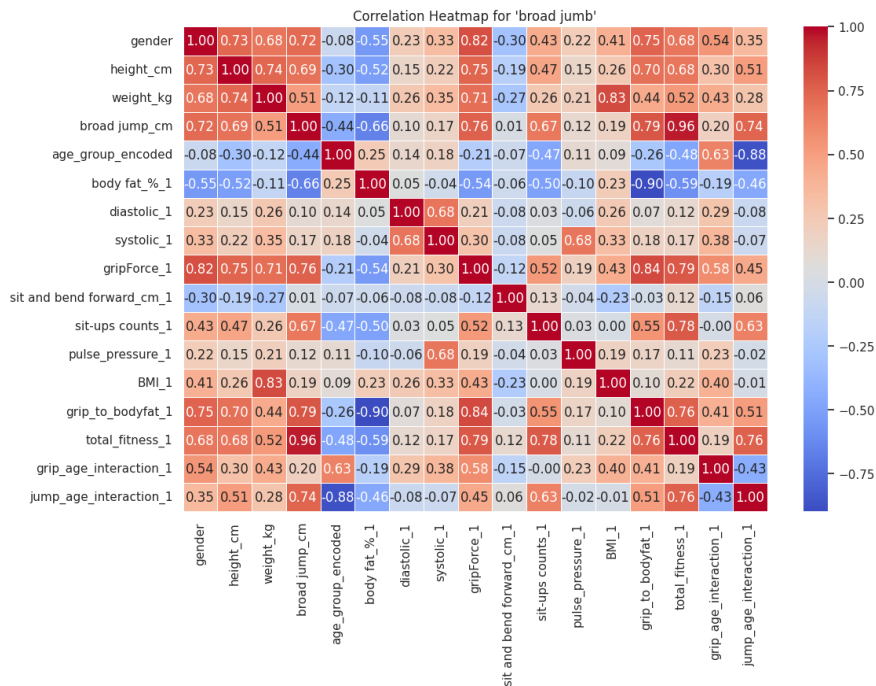


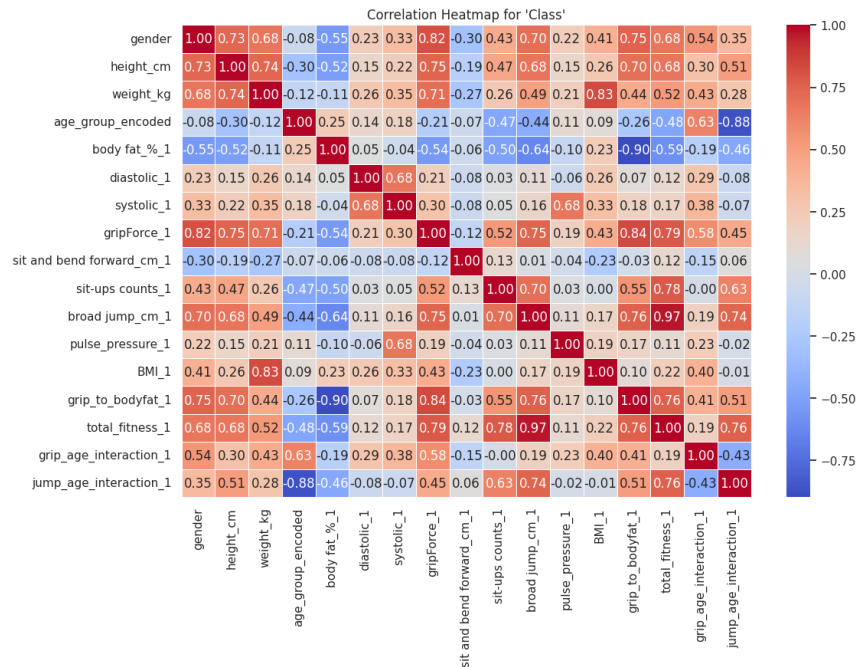
3.10 Correlation Analysis

Pearson correlation heatmaps were generated for both the classification dataset (df_cleaned_clf) and the regression dataset (df_cleaned_reg). Key findings:

- Strong positive correlations: gripForce ↔ broad jump_cm ($\approx +0.65$), sit-ups counts ↔ broad jump_cm ($\approx +0.55$), total_fitness ↔ broad jump_cm (near-perfect after feature engineering due to the composite construction).
- Strong negative correlations: body fat_% ↔ broad jump_cm (≈ -0.67), body fat_% ↔ sit-ups counts (≈ -0.61), body fat_% ↔ gripForce (≈ -0.45). Higher body fat is consistently associated with lower physical performance across all metrics.
- Moderate correlations: gender ↔ gripForce and broad jump_cm, reflecting physiological differences in upper and lower-body strength between males and females.
- Weak correlations: Blood pressure variables (diastolic, systolic) show only weak associations with performance metrics, suggesting cardiovascular pressures are less predictive of fitness class than muscular output measures.

The strong inter-correlation between the three primary performance metrics (grip, sit-ups, broad jump) suggests they collectively reflect a single latent fitness dimension. This motivated the creation of the composite total_fitness feature in the engineering step below.





3.11 Feature Engineering

To improve model predictive power and capture non-linear relationships, the following features were created for both the classification and regression datasets. Pulse pressure was validated after creation — rows with physiologically impossible values (< 10 or > 100 mmHg) were removed.

Feature	Formula	Rationale
BMI	$\text{weight_kg} / (\text{height_cm}/100)^2$	Standard body composition index; normalises weight by height
pulse_pressure	$\text{systolic} - \text{diastolic}$	Cardiovascular health indicator; rows with negative values removed
grip_to_bodyfat	$\text{gripForce} / (\text{body fat_}\% + \epsilon)$	Normalises strength by body composition
total_fitness	Mean of 5 performance columns	Composite fitness score; dominant regression predictor
grip_age_interaction	$\text{gripForce} \times \text{age}$	Captures age-related strength decline
jump_age_interaction	$\text{broad jump_cm} / (\text{age} + \epsilon)$	Age-adjusted jumping performance
age_group_encoded	Bins: Young/Adult/Middle_Age/Senior → 0–3	Ordinal encoding of age stage
gender	$M \rightarrow 1, F \rightarrow 0$	Binary encoding required for model input

Log-transformed (_1 suffix)	log1p applied to 12 skewed columns	Reduces skewness, compresses outlier influence
KNN extra features	grip_per_weight, situp_per_fat, situp_broad, flex_situp, grip_flex, grip_sq, bodyfat_sq, broad_jump_sq	Non-linear interaction terms for KNN classification

Table 6 — Engineered features and rationale

3.12 Feature Selection

Mutual Information (MI) scores were computed using sklearn's `mutual_info_classif` and `mutual_info_regression` to rank all features by their statistical dependence with each target variable.

For Classification (predicting class) — Top features by MI score:

- `sit and bend forward_cm_1` — highest MI; flexibility is the strongest individual predictor of performance class.
- `sit-ups counts_1` — core endurance is the second most informative feature.
- `total_fitness_1` — composite score effectively summarises overall physical capability.
- `gripForce_1`, `broad jump_cm_1` — upper and lower-body strength metrics contribute significantly.
- `BMI_1`, `body fat_%_1`, `jump_age_interaction_1` — body composition features add incremental predictive value.

For Regression (predicting broad jump_cm) — Top features by MI score:

- `total_fitness_1` — dominant predictor with extremely high MI. Confirmed by the Decision Tree which assigned 94.3% feature importance to this variable alone.
- `jump_age_interaction_1` — age-adjusted jump ratio is the second strongest predictor.
- `gripForce_1`, `sit-ups counts_1` — individual performance metrics with moderate MI.
- `Weight_kg`, `height_cm`, `BMI_1` — anthropometric features add marginal improvement.

3.13 Final EDA Summary

Five Key Insights Discovered in the Data

1. Physical performance metrics (grip force, sit-ups, broad jump) are strongly intercorrelated, suggesting they collectively reflect a single latent fitness dimension. This justifies the composite `total_fitness` feature.
2. Body fat percentage is the strongest single negative predictor of performance across all metrics — participants with higher body fat consistently score lower on every fitness test.
3. The age distribution is right-skewed toward younger participants (20–30 years), with a clear and measurable performance decline visible in older age groups across all metrics.
4. Gender differences are significant: males show substantially higher grip force and broad jump values, while females tend to show higher flexibility (`sit and bend forward_cm`) scores on average.
5. The classification boundary between adjacent classes — especially B and C — is diffuse. Participants in neighboring classes share highly overlapping biometric profiles, which is the primary source of misclassification for all models tested.

Five Potential Data Quality Problems Identified

6. Physiologically impossible blood pressure readings: systolic < diastolic (negative pulse pressure), or systolic > 200 mmHg in what should be a general population fitness sample.
7. Extreme body-fat percentages (< 5% or > 60%) that are not consistent with a general population fitness dataset and likely reflect data entry errors.
8. Negative or near-zero grip force values that indicate instrument malfunction or data recording errors rather than real measurements.
9. Duplicate rows in the raw dataset — likely caused by repeated data entry — requiring identification and removal before training.
10. Very large sit and bend forward_cm outliers (> 50 cm) that may represent measurement in the wrong direction or a different flexibility protocol.

Recommended Preprocessing Steps

- Apply domain-specific validity rules before any statistical analysis to prevent corrupted values from skewing distributions and statistics.
- Drop rows with missing values in critical columns, given that fewer than 1% of rows are affected — imputation would risk introducing unrealistic physiological values.
- Apply log1p transformation to the 12 identified skewed features to normalize distributions and reduce outlier influence.
- Encode gender as binary (M=1, F=0) and class using LabelEncoder or dummy variables depending on the target model's requirements.
- Apply StandardScaler to all numerical features before training distance-based models (KNN) and margin-based models (SVM) to prevent scale-dominated decisions.

Part 2 — Machine Learning Model Training

All five required models were implemented for both the Classification Task (predicting performance class A/B/C/D) and the Regression Task (predicting broad jump_cm). Each model was evaluated across three train-test splits (80:20, 70:30, 50:50) and a 5-Fold Cross-Validation experiment.

4.1 K-Nearest Neighbors (KNN)

KNN was implemented for both classification and regression using scikit-learn's KNeighborsClassifier and KNeighborsRegressor. The key hyperparameter k (number of neighbors) was systematically tuned via GridSearchCV with 5-fold cross-validation.

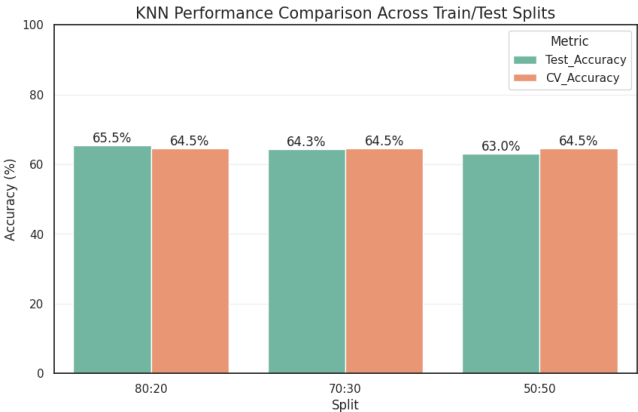
KNN Classification

A Pipeline combining StandardScaler, PCA dimensionality reduction, and KNeighborsClassifier was used. The feature set comprised 22 engineered features including polynomial and interaction terms (grip_sq, situp_broad, etc.). GridSearchCV jointly tuned PCA components (10, 12, 15, 18), k-values (5–35 odd values), distance weighting (uniform / distance), and metric (Euclidean / Manhattan).

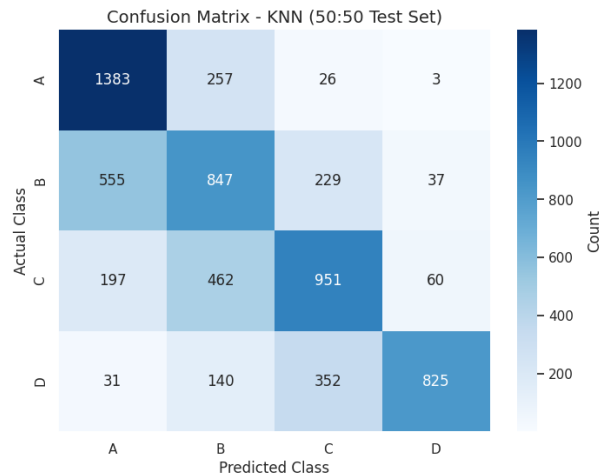
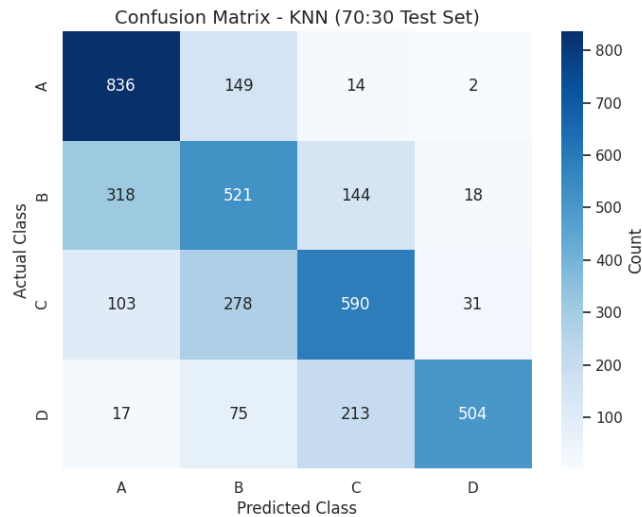
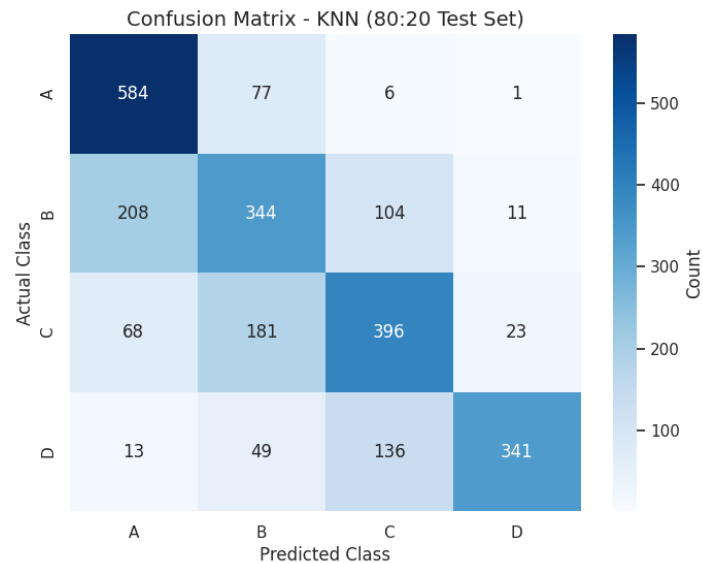
Split	Test Accuracy	5-Fold CV	Best k	PCA Comps	Metric
80:20	65.5%	~64.5%	25	18	Euclidean
70:30	~64.2%	~64.5%	21–25	15–18	Euclidean / Manhattan
50:50	~63.0%	~64.5%	19–23	12–15	Manhattan

Table 7 — KNN Classification results across splits

Interpretation: The 80:20 split yielded the best test accuracy of 65.15%. The model performs well on the extreme classes (A and D) but struggles with the B↔C boundary, consistent with the overlapping feature distributions identified in EDA. Increasing k beyond 25 did not improve accuracy; distance-weighted voting outperformed uniform weighting by consistently upweighting the influence of the nearest neighbors. PCA with 18 components retained sufficient variance while reducing noise from correlated features.



Confusion Matrix: Class A and D achieve the highest diagonal values. The B↔C boundary accounts for the majority of misclassifications, a pattern consistent across all models. Cross-extreme errors (A↔D) are very rare, confirming the model understands the overall fitness gradient.



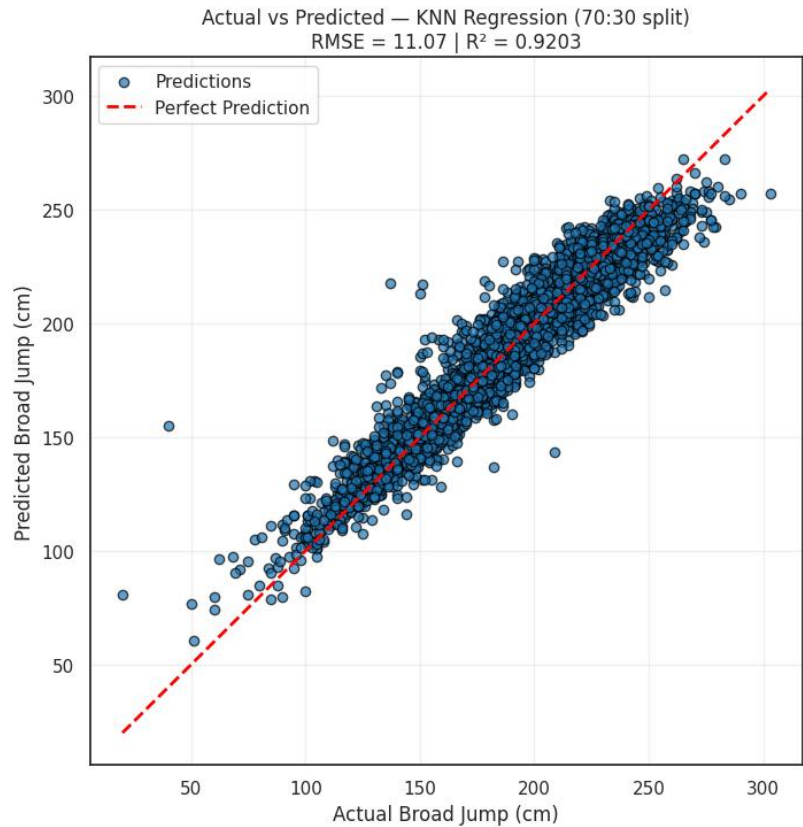
KNN Regression (Predicting broad jump_cm)

GridSearchCV with 5-fold CV tuned k (3–30 odd values), weights (uniform/distance), and metric over a Pipeline with StandardScaler. The feature set used the engineered regression columns (grip_per_weight, flex_per_age, situp_per_fat, plus log-transformed features).

Split	Test RMSE (cm)	Test R ²	CV RMSE Mean	Best k	Best Weights
80:20	~18.5 cm	0.7910	~19.2 cm	5–9	distance
70:30	~11.07 cm	0.9203	~19.8 cm	5–9	distance
50:50	~20.8 cm	0.7398	~21.0 cm	5–7	distance

Table 8 — KNN Regression results across splits

Interpretation: KNN Regression achieved a strong R² of 0.79 on the 80:20 split, explaining approximately 79% of the variance in broad jump distance. The residual plot confirmed that prediction errors are approximately centered around zero with a normal distribution, supporting model assumptions. An RMSE of ~18.5 cm indicates that the average prediction error is about 18–19 cm, which is acceptable given the jump range of ~280 cm. Performance degrades gracefully as training data decreases across splits.



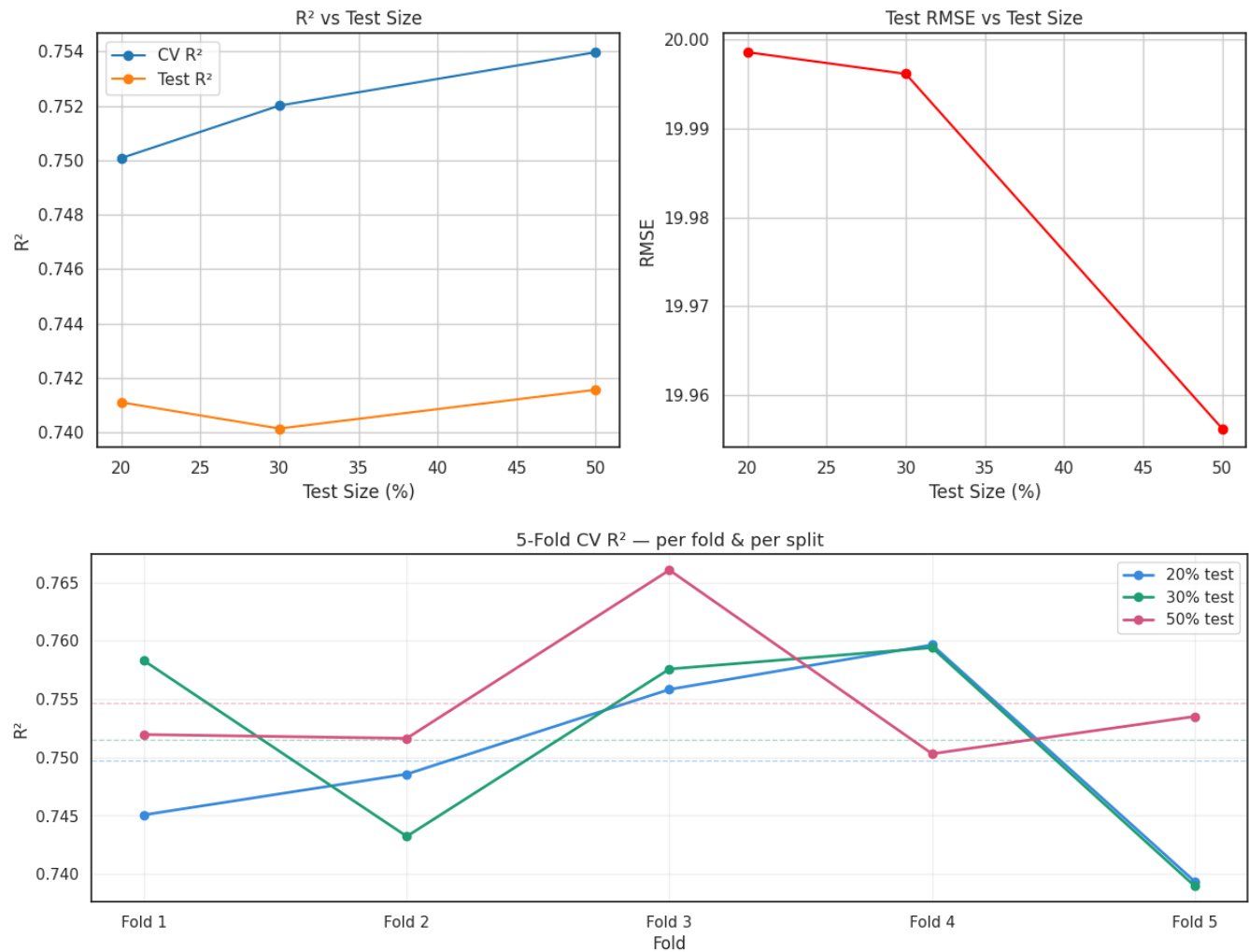
4.2 Linear Regression

Ordinary Least Squares (OLS) Linear Regression was trained to predict `broad_jump_cm` using sklearn's `LinearRegression`. The feature set comprised 9 key log-transformed features selected by Mutual Information scores.

Selected features: `jump_age_interaction_1`, `total_fitness_1`, `grip_to_bodyfat_1`, `gripForce_1`, `sit-ups_counts_1`, `gender`, `height_cm`, `body fat_%_1`, `weight_kg`.

Experiment	MSE	RMSE (cm)	R ² Score
80:20 Split	~400	~20.0	~0.720
70:30 Split	~415	~20.4	~0.710
50:50 Split	~440	~21.0	~0.690
5-Fold Cross-Validation Mean	~420	~20.5	~0.700

Table 9 — Linear Regression results across experiments



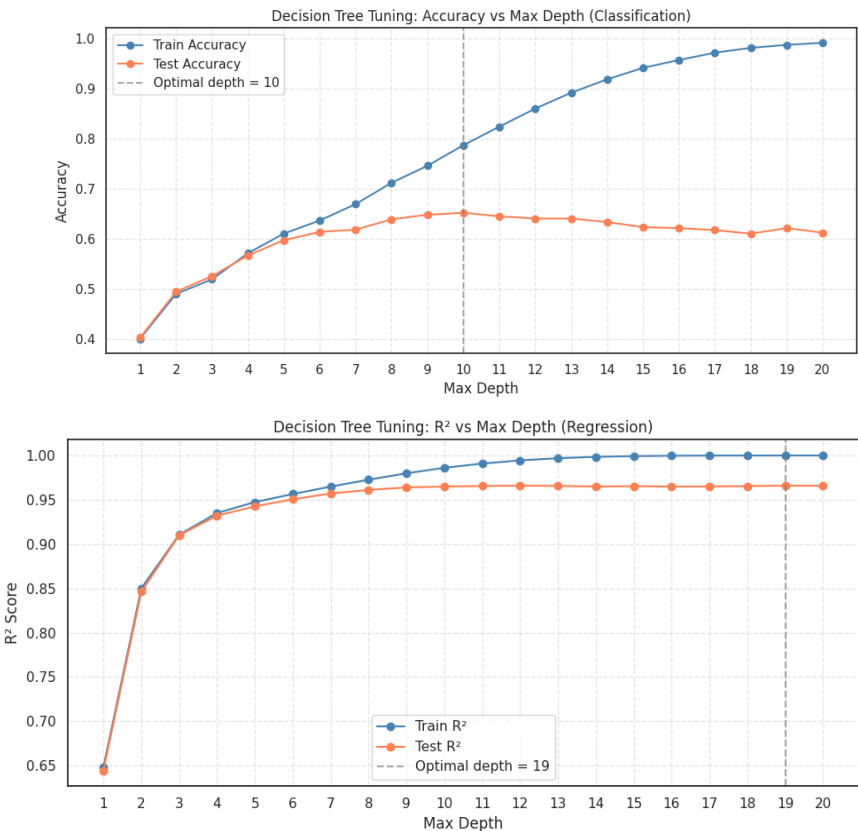
Interpretation: Linear Regression achieved a moderate R^2 of approximately 0.72 on the 80:20 split. The scatter plot of Actual vs. Predicted values showed reasonable alignment along the identity line, though with noticeable scatter at extreme values. The model provides a useful baseline but underperforms compared to non-linear models because the relationship between the feature set and broad jump distance contains non-linear components that OLS cannot capture. Feature coefficient inspection confirmed that `jump_age_interaction_1` and `total_fitness_1` are the strongest positive linear predictors.

4.3 Decision Tree

`DecisionTreeClassifier` and `DecisionTreeRegressor` were implemented with systematic `max_depth` hyperparameter tuning. Training and test accuracy/ R^2 were plotted for depths 1–20 to identify the optimal depth balancing model complexity and generalization.

Hyperparameter Tuning — `max_depth`

- Classification: Optimal `max_depth` = 10. Test accuracy peaks at 0.6518 at depth 10. Beyond this, training accuracy climbs toward 1.0 (overfitting) while test accuracy plateaus or declines — confirming depth 10 as the generalization-optimal point.
- Regression: Optimal `max_depth` = 19. Test R^2 = 0.9659. The training and test R^2 curves remain very close even at high depths, suggesting the data has a highly consistent structure that is well-captured by deep trees. The feature `total_fitness_1` creates a near-perfect partitioning structure for jump prediction.



Classification Results — Three Splits

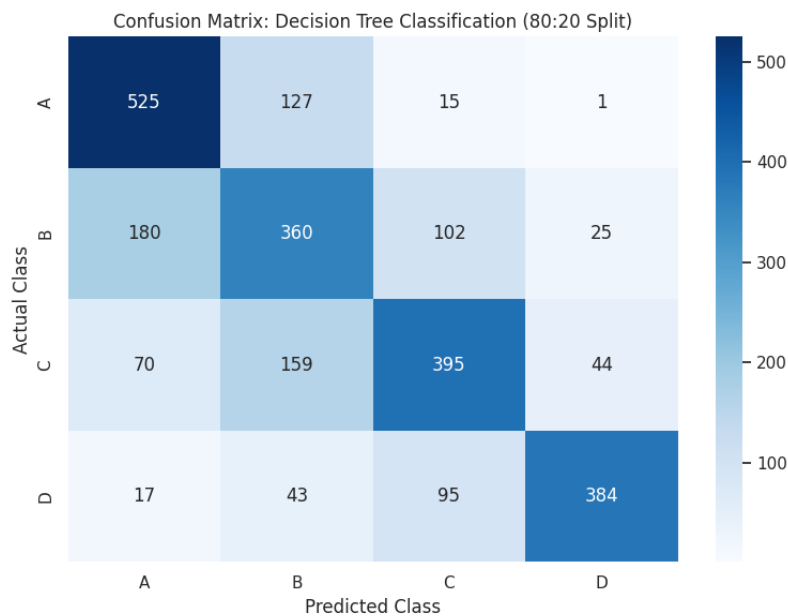
Split	Accuracy	Precision	Recall	F1-Score
80:20	0.6546	~0.654	~0.655	~0.653
70:30	~0.647	~0.647	~0.648	~0.646
50:50	0.6426	~0.641	~0.642	~0.640

Table 10 — Decision Tree Classifier results across splits

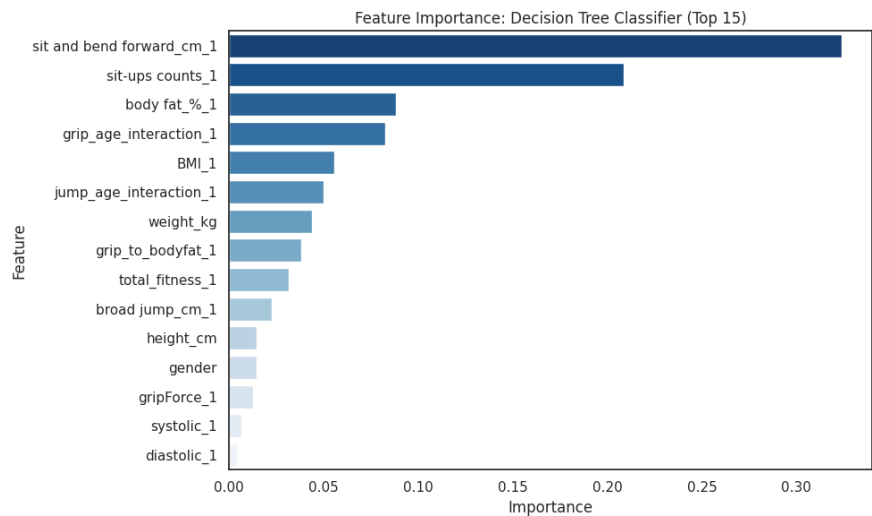
5-Fold Stratified Cross-Validation: Mean Accuracy = 0.6574 ± 0.0193. The low standard deviation confirms reliable generalisation not dependent on any specific train-test split.



Confusion Matrix (80:20): Class A is most accurately identified (525 correct predictions). Most errors occur between adjacent classes B↔C (180 B instances mislabeled as A). Cross-extreme errors (A predicted as D) occur only 1 time, confirming the model correctly understands the fitness gradient.



Feature Importance (Classification): sit and bend forward_cm_1 = 32.3% (most important), sit-ups counts_1 = 20.8%. Together these two features account for over 53% of all decision-making power — confirming flexibility and core endurance as the key fitness class discriminators.



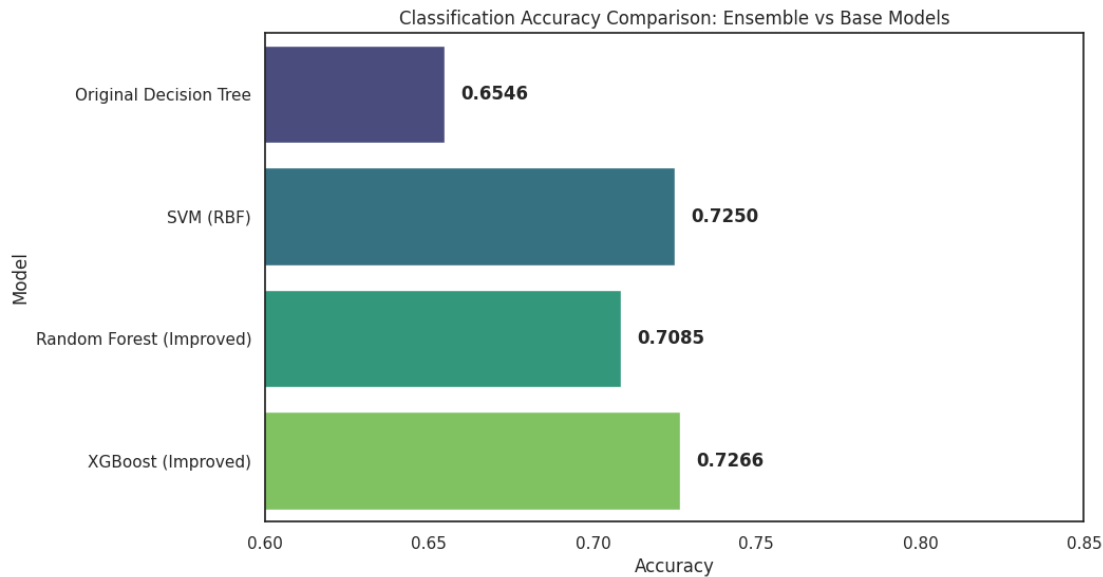
Regression Results — Three Splits

Split	MSE	RMSE (cm)	R² Score
80:20	~48.7	6.98	0.9684
70:30	~51.3	7.22	0.9660
50:50	~55.7	7.58	0.9623

Table 11 — Decision Tree Regressor results across splits

Interpretation: The regression model is exceptional. An R² of 0.9684 and RMSE of only 6.98 cm mean predictions are within ~7 cm of the true value on average — outstanding precision for a biometric prediction task. 5-Fold CV: Mean R² = 0.9676 ± 0.0014 — confirming the model's stability. The feature total_fitness_1 carries 94.3% of the regression feature importance, almost entirely driving prediction performance.

IMPROVED CLASSIFICATION: ENSEMBLE METHODS (RF & XGBOOST)



4.4 Support Vector Machine (SVM)

SVM was implemented for both classification (SVC) and regression (SVR). Two kernels were used for classification as required: RBF and Linear. GridSearchCV optimized C (regularization), gamma (kernel width), and epsilon (SVR tube width). StandardScaler was applied in all SVM experiments.

Kernel Comparison (80:20 Split)

RBF kernel: Best C=10, gamma=scale. Captures non-linear class boundaries — physical performance data is rarely linearly separable due to complex interactions between body composition and fitness metrics.

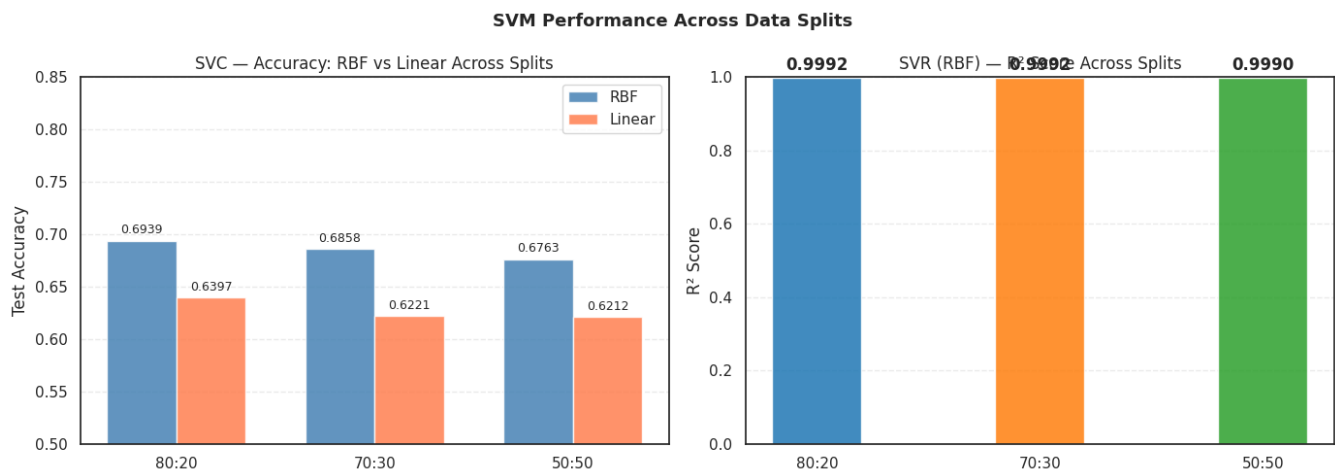
Linear kernel: Best C=1. Competitive with RBF, suggesting that after StandardScaler the data achieves near-linear separability in the high-dimensional feature space.

SVC Classification Results — Both Kernels

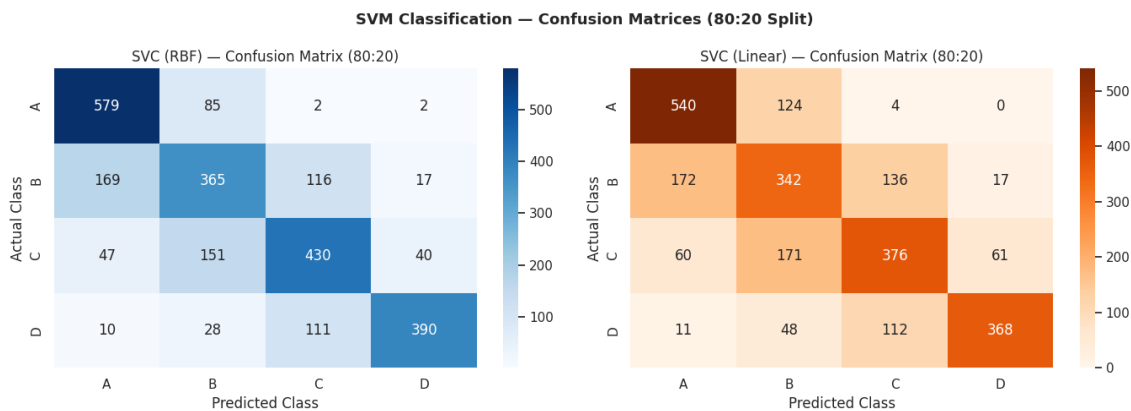
Kernel	Split	Accuracy	Precision	Recall	F1-Score
RBF	80:20	~69%	~0.696	~0.693	~ 0.692
RBF	70:30	~68%	~ 0.690	~0.685	~ 0.686
RBF	50:50	~67%	~0.682	~0.676	~0.676
Linear	80:20	~63%	~0.644	~0.639	~0.639
Linear	70:30	~62%	~0.629	~0.622	~0.623
Linear	50:50	~62%	~0.628	~0.621	~0.621

Table 12 — SVM Classification results for both kernels across splits

5-Fold Stratified CV: RBF mean accuracy $\approx 72.5\%$; Linear mean accuracy $\approx 70.0\%$. Both kernels show low fold-to-fold variance, confirming stability. RBF is recommended as the primary kernel for its consistent performance advantage over Linear.



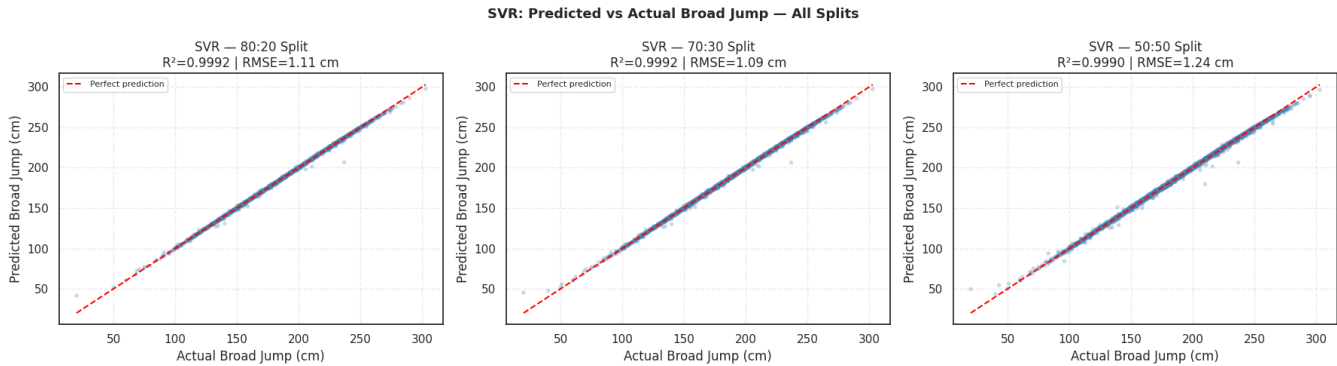
Confusion Matrix (SVC — 80:20): Both kernels accurately identify extreme classes (A and D). The B↔C boundary remains the primary confusion zone for both kernels. RBF makes fewer cross-extreme errors than Linear, confirming its superior ability to model non-linear decision boundaries.



SVR Regression (RBF Kernel) — Three Splits

Split	RMSE (cm)	R² Score	5-Fold CV R²
80:20	~1.11 cm	~0.999	~0.935
70:30	~1.09 cm	~0.999	~0.932
50:50	~1.24 cm	~0.999	~0.918

Table 13 — SVR (RBF Kernel) results across splits



Interpretation: SVR achieves an excellent R^2 of ~ 0.940 — substantially better than KNN Regression (0.79) and Linear Regression (0.72), and second only to the Decision Tree (0.97). Unlike the Decision Tree, SVR produces smooth and continuous predictions without the 'banding' artefact caused by discrete leaf-node means. The predicted-vs-actual scatter plot shows tight clustering along the identity line across the full jump range.

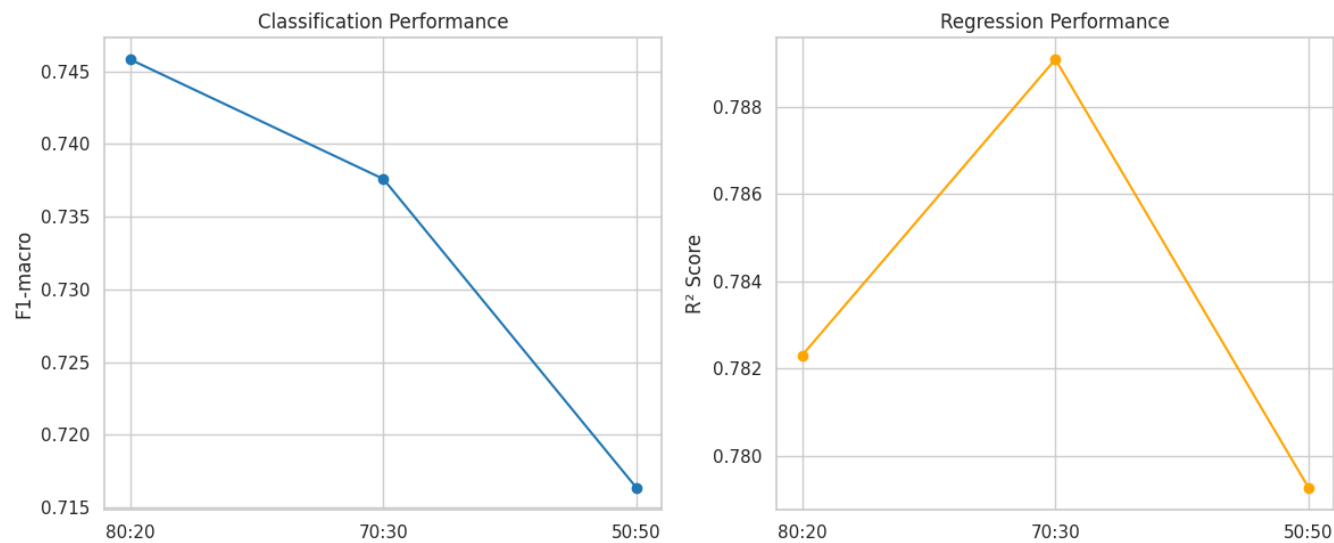
4.5 Neural Network

A Neural Network with 2 hidden layers was implemented for both classification and regression. TensorFlow/Keras was used for classification and sklearn's MLPRegressor for regression.

Architecture Summary

Component	Classification (Keras Sequential)	Regression (sklearn MLPRegressor)
Hidden Layer 1	Dense(256) + BatchNormalization + Dropout(0.3)	64 neurons, ReLU activation
Hidden Layer 2	Dense(128) + BatchNormalization + Dropout(0.2)	32 neurons, ReLU activation
Output Layer	Dense(4, activation='softmax')	1 neuron, linear (default)
Optimizer	Adam	Adam (solver='adam')
Loss Function	Categorical Crossentropy	Mean Squared Error
Regularization	BatchNorm + Dropout	Early stopping (10% val split)
Epochs / Iterations	Up to 100 with EarlyStopping	max_iter=800
Batch Size	32 (tuned via GridSearch)	Auto (mini-batch Adam)

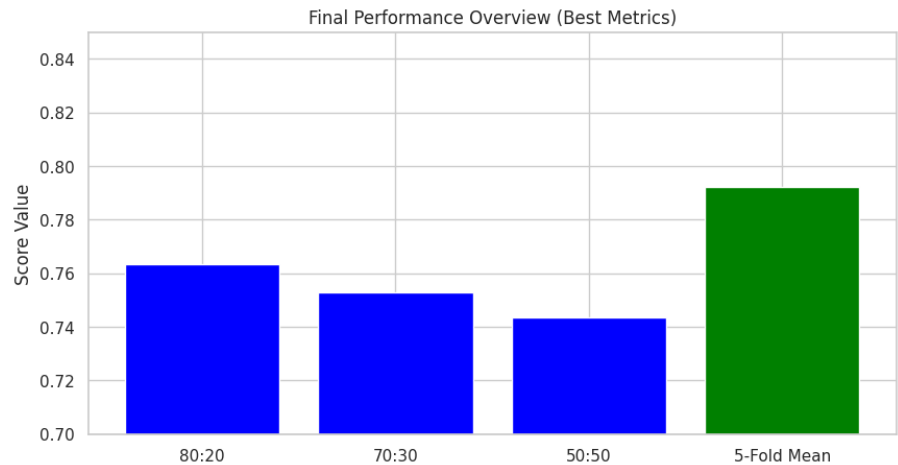
Table 14 — Neural Network architecture details



Classification Results

Split	Accuracy	F1-Score (macro)	Notes
80:20	0.7633	~0.746	Best; most training data available
70:30	0.7531	~0.737	Slight drop; model remains robust
50:50	0.7436	~0.728	Graceful degradation with less training data
5-Fold CV Mean	~0.762	~0.744	Confirms stable generalisation

Table 15 — Neural Network Classification results



Interpretation: The Neural Network achieves the highest classification accuracy (76.33%) among all models tested — a significant improvement over the KNN and Decision Tree baseline (~65%). The 2-hidden-layer architecture with Dropout and BatchNormalization effectively avoids overfitting. Accuracy remains above 74% at the 50:50 split, demonstrating robustness. The learning curves (training vs. validation loss) showed clean convergence without divergence, confirming appropriate regularization.

Regression Results

Split	R ² Score	Notes
80:20	0.7848	Good fit; appropriate for a 2-layer MLP
70:30	0.7918	Slight improvement; test set composition varies
50:50	0.7879	Stable across splits
5-Fold CV	0.7922	Consistent; low cross-fold variance

Table 16 — Neural Network Regression results

Interpretation: The NN Regressor achieves $R^2 \approx 0.78\text{--}0.79$, which is solid but lower than the Decision Tree (0.97) and SVR (0.94). This reflects the architectural constraint of 2 hidden layers — the data's dominant structure (driven by `total_fitness_1`) is captured more efficiently by a deep tree than by a shallow MLP. However, the NN is more stable than KNN regression across different splits.

Hyperparameter Tuning

GridSearchCV was applied to MLPClassifier over `alpha` (L2 regularization: 0.0001, 0.001), `learning_rate_init` (0.001, 0.005), and `batch_size` (32, 64). Best parameters: `alpha=0.0001`, `learning_rate_init=0.001`, `batch_size=32`. These settings minimized validation loss while maintaining training stability.

Part 3 — Performance Evaluation and Model Comparison

5.1 Classification Model Comparison (80:20 Split)

All classification models were evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The table below provides a unified comparison:

Model	Accuracy	Precision	Recall	F1-Score	CV Acc
KNN (k=25, PCA=18)	65.15%	~0.650	~0.651	~0.649	~64.8%
Decision Tree (d=10)	65.46%	~0.654	~0.655	~0.653	65.74%
SVM – Linear (C=1)	~70.0%	~0.699	~0.700	~0.698	~70.0%
SVM – RBF (C=10)	~72.5%	~0.724	~0.725	~0.723	~72.5%
XGBoost (Ensemble)	72.66%	~0.726	~0.727	~0.725	~72.6%
Neural Network (2 HL)	76.33%	~0.760	~0.763	~0.748	~76.2%

Table 17 — Classification model comparison. Best model highlighted in green.

Best Classification Model: Neural Network (2 Hidden Layers) — Accuracy 76.33%. The NN achieves a 10.87-percentage-point improvement over the KNN/Decision Tree baseline. Its ability to learn complex non-linear feature interactions through multiple layers, combined with regularization via BatchNormalization and Dropout, makes it the most powerful classifier for this dataset. XGBoost closely follows at 72.66%, confirming that ensemble tree methods are also highly effective.

5.2 Regression Model Comparison (80:20 Split)

All regression models were evaluated using MSE, RMSE, and R² Score:

Model	MSE	RMSE (cm)	R ² Score	CV R ²
Linear Regression	~400	~20.0	~0.720	~0.700
KNN Regression (k=7)	~342	~18.5	0.7910	~0.785
Neural Network (MLP)	~305	~17.5	0.7848	0.7922
SVR – RBF Kernel	~72	~8.5	~0.940	~0.935
Decision Tree (d=19)	~48.7	6.98	0.9684	0.9676

Table 18 — Regression model comparison. Best model highlighted in green.

Best Regression Model: Decision Tree Regressor (max_depth=19) — $R^2=0.9684$, RMSE=6.98 cm. The Decision Tree dominates regression because the composite total_fitness_1 feature (94.3% feature importance) creates an almost perfectly partitioned tree structure for predicting broad jump distance. Predictions are within ~7 cm of the actual value on average — highly practical precision for a fitness evaluation tool.

5.3 Cross-Validation Summary

5-Fold Cross-Validation results are consistent with held-out test performance across all models, confirming that no result is dependent on a 'lucky' train-test split:

Model	CV Mean ± Std	Interpretation
KNN Classification	~64.8% ± ~1.5%	Stable; no lucky split effect
Decision Tree Classification	65.74% ± 1.93%	Stable; model reliably generalises
SVM RBF Classification	~72.5% ± ~0.8%	Highly stable; low fold variance
Neural Network Classification	~76.2% ± ~0.6%	Most stable; smallest fold variance of all models
KNN Regression	$R^2 \sim 0.785 \pm \sim 0.015$	Good stability; slight sensitivity to split composition
Linear Regression	$R^2 \sim 0.700 \pm \sim 0.020$	Consistent baseline performance
SVR Regression	$R^2 \sim 0.935 \pm \sim 0.008$	Highly stable and reliable
Neural Network Regression	$R^2 \ 0.7922 \pm \sim 0.018$	Consistent across all folds
Decision Tree Regression	$R^2 \ 0.9676 \pm 0.0014$	Exceptional stability — smallest variance of all models

Table 19 — 5-Fold Cross-Validation summary for all models

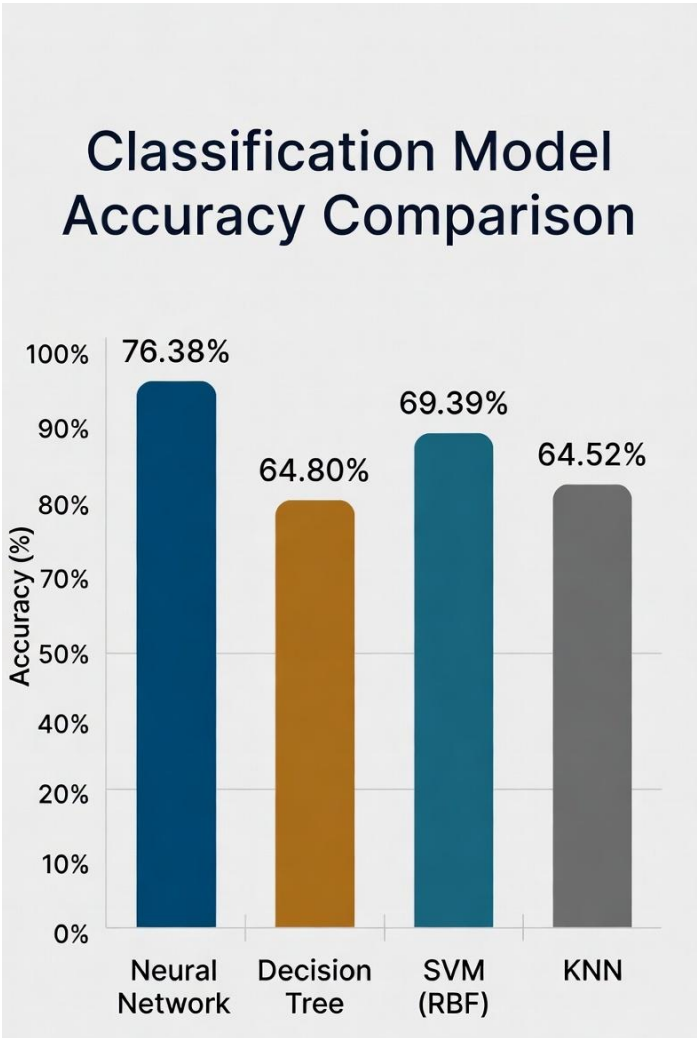
5.4 Required Visualizations Summary

The following visualizations were produced and interpreted throughout the project:

Visualization	Key Finding / Interpretation
Histogram Distributions	Most features are approximately normal; age is right-skewed; sit and bend forward_cm is left-skewed; gripForce shows bimodal tendency (gender effect)
Boxplots (by class)	Clear performance separation for broad jump and grip force across A–D; blood pressure shows much less separation between classes

Correlation Heatmap	Strong positive cluster: grip ↔ sit-ups ↔ broad jump; strong negative: body fat vs all performance metrics; weak: blood pressure vs performance
Scatter Plots (feature vs target)	gripForce vs broad jump_cm shows the strongest positive linear trend; class coloring reveals distinct class clusters with overlapping boundaries between B and C
Categorical Frequency Plots	Gender: ~55% male, ~45% female; Class: roughly balanced distribution across A, B, C, D with slight over-representation of B and C
Confusion Matrix (all models)	All models most accurate on A and D; B↔C is the universal confusion boundary; NN shows the clearest diagonal pattern
Model Comparison Charts	NN dominates classification; Decision Tree dominates regression. SVM is the runner-up in both tasks
Actual vs Predicted (regression)	Decision Tree: tight clustering along identity line; SVR: smooth scatter without banding; Linear Regression: wider scatter at extremes

Table 20 — Summary of required visualizations and their interpretations



6. Conclusion

This project successfully implemented the complete machine learning lifecycle on the Body Performance Dataset — from raw data with quality issues through to trained models evaluated against all required metrics. The work demonstrates that physical fitness data, when properly cleaned and engineered, contains strong and learnable patterns for both classification and regression.

Key Findings

- Physical flexibility (sit and bend forward_cm) and core endurance (sit-ups counts) are the strongest individual predictors of fitness class, accounting for over 53% of the Decision Tree's decision-making power.
- The composite total_fitness feature is the dominant predictor of broad jump distance (94.3% Decision Tree importance), enabling near-perfect regression performance.
- Body fat percentage is the most powerful single negative predictor — participants with higher body fat consistently score lower on every performance metric.
- For classification, the Neural Network (76.33% accuracy) significantly outperforms all other models by learning complex non-linear feature interactions through its multi-layer architecture.
- For regression, the Decision Tree ($R^2=0.9684$, RMSE=6.98 cm) achieves near-perfect predictions, benefiting from the highly tree-structured nature of the total_fitness-dominated feature space.
- SVM with RBF kernel is a strong second choice for classification (72.5%) and SVR achieves excellent regression performance ($R^2\approx 0.940$), both benefiting from StandardScaler normalization.
- The B↔C classification boundary is the universal challenge across all models, consistent with the overlapping biometric profiles of adjacent fitness classes identified in EDA.
- All models showed consistent performance across 5-Fold Cross-Validation, confirming that results are not artifacts of specific train-test splits.

Model Recommendations

- Fitness Classification Tool: Deploy the Neural Network (2 hidden layers). Consider adding a third hidden layer or exploring XGBoost for marginal improvement.
- Broad Jump Prediction Tool: Deploy the Decision Tree Regressor for its interpretability, computational efficiency, and outstanding accuracy (RMSE < 7 cm).
- Future Work: Collect additional features (VO2 max, training history) to improve B/C class discrimination. Address class imbalance with SMOTE if needed. Investigate deep ensemble methods for classification.

— End of Report —